

First name:

Last name:

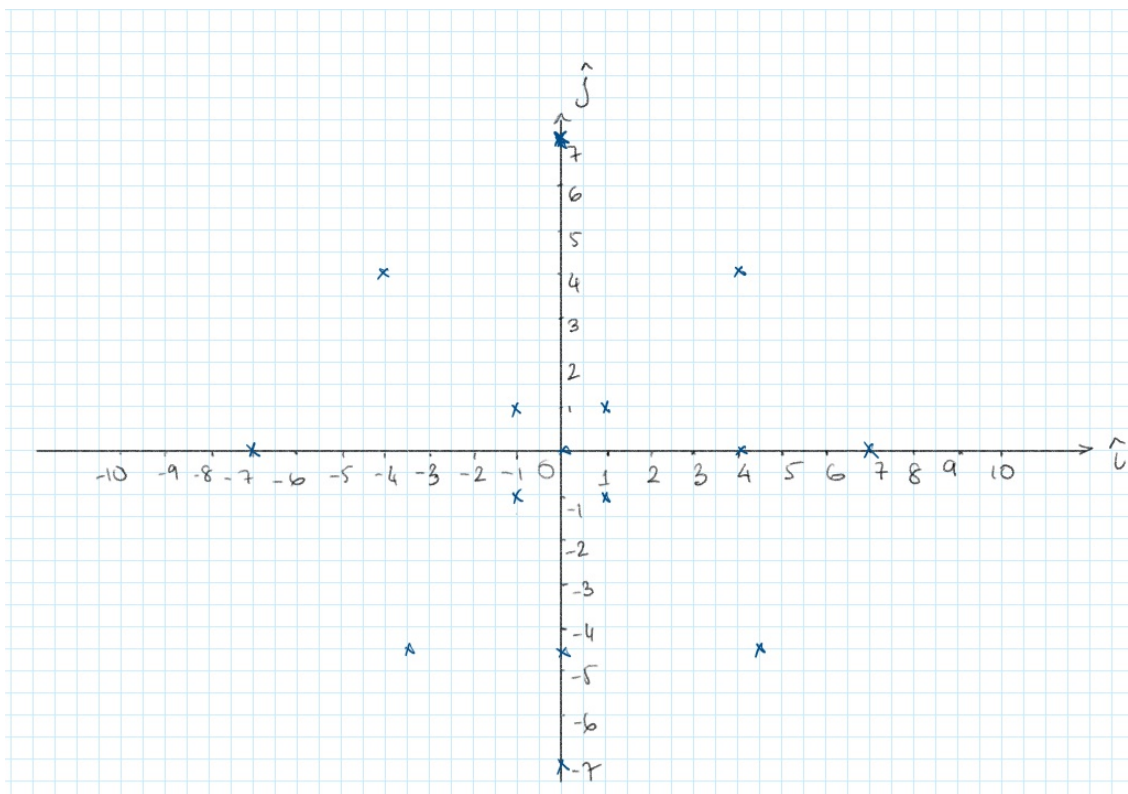
Student ID:

Problem 1. Sketch vector arrows at the points marked with 'x' to depict the vector field

$$\vec{v} = -y\hat{i} + x\hat{j}.$$

[5 marks]

Solution.



□

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Problem 2. Find the first and second derivatives (with respect to t) of

$$\vec{r}(t) = \begin{bmatrix} 3 \cos(2t) \\ 3 \sin(2t) \\ 4t \end{bmatrix}.$$

[5 marks]

Solution.

□

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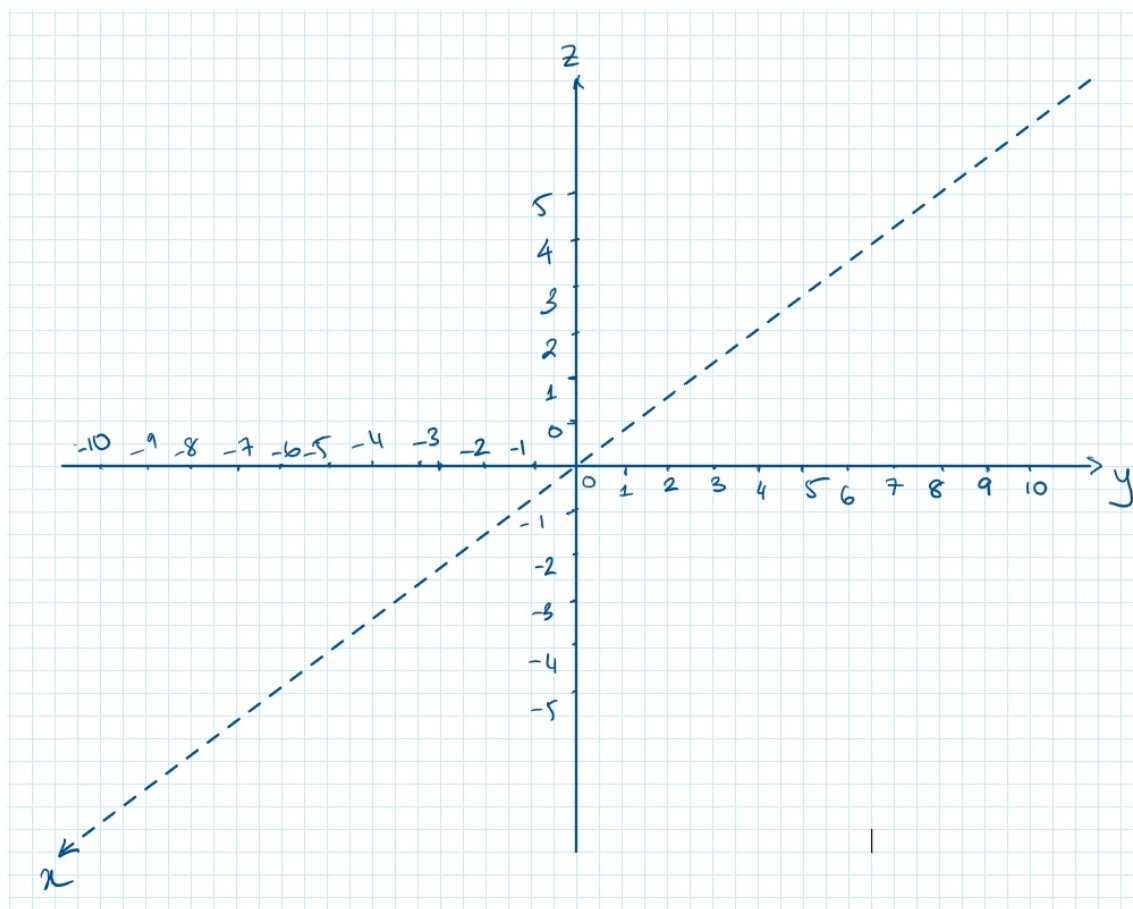
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Problem 3. Find a parametric representation of a circle in the yz -plane, with center $(0, 4, 0)$ and passing through $(0, 0, 3)$. Sketch the curve on the given axes.

[5 marks]

Solution. Parametric representation:



Sketch:

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